

C104J



SEAT No. _____

Number of pages: 02

SARDAR PATEL UNIVERSITY

Master of Science (Information Technology) – FIRST SEMESTER

PS01CINT54 - (Relational Database Management Systems)

15th February, 2022, Tuesday

Time: 12 PM To 2 PM

Marks: 70

Q1(A) Choose the most appropriate option for each question:

[8]

1. _____ is called as distinct piece of information.
 (A) Data (B) Information
 (C) Record (D) None of given
2. ERD stands for _____.
 (A) Entity Relationship Diagram (B) Entity Referential Diagram
 (C) Entity Representation Diagram (D) None of given
3. _____ is the process of restoring a database to the correct state in the event of a failure.
 (A) Data Sharing (B) Data Integrity
 (C) Data Recovery (D) None of given
4. In ER diagram, _____ are represented by ellipse.
 (A) Relationships (B) Attributes
 (C) Entity (D) All of given
5. _____ is an output of MOD(5,2).
 (A) 2 (B) 1
 (C) 0 (D) None of given
6. _____ is a Conditional control structure.
 (A) FOR loop (B) GOTO statement (C) IF statement (D) None of given
7. _____ returns always one value.
 (A) Function (B) Procedure
 (C) Trigger (D) Function & Procedure both
8. _____ is executed automatically.
 (A) Function (B) Trigger
 (C) Procedure (D) Procedure & Trigger both

(B) Do as directed.

[16]

1. _____ in Database Management System is a procedure of managing simultaneous operations without conflicting with each other.
2. Which symbol is used to represent Entity in ER diagram?.
3. Multi-valued attributes are represented by _____ symbol in ERD.
4. _____ is a group of tables and views that incorporate vital details regarding a database.
5. When only one instance of an entity is associated with the one instance of another entity then that relationship is called _____.
6. The ability to share the same data resource with multiple applications or users is known as _____.
7. _____ are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data.
8. _____ command is used to change values of records in a table.
9. Oracle provides an object _____ that can generate numerical values.
10. TRUNC(125.815,-2) returns _____.
11. Which clause is used to impose condition on output given by GROUP BY clause?
12. A _____ is a database object that holds user data.

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13. _____ constraint establishes uniqueness of records in a table
14. Which section is used to handle errors in PL/SQL?
15. The data stored in the cursor is called _____.
16. In PL/SQL data block there are maximum _____ sections.
- Q2 Answer the following questions (ANY SEVEN): [14]**
1. Write the purpose of primary key and foreign key.
 2. Define ER Diagram.
 3. Differentiate clearly Physical data independence & Logical data independence.
 4. List any four data types.
 5. Write a SQL command to display result of 500 * 600.
 6. Explain briefly IF statement.
 7. Differentiate clearly Trigger and Stored procedure
 8. Write SQL command to remove primary key SNO of a table STUD.
 9. Explain with example the built in function CEIL().
- Q-3 Define DBMS and write the advantages of using DBMS approach. [8]**
- OR**
- Q3 Explain briefly with a diagram the different categories of data models. [8]**
- Q4 Write short note on types of attributes. [8]**
- OR**
- Q4 What is the purpose of Normalization? Explain briefly the process of normalization by taking appropriate example(s). [8]**
- Q5 Do as directed: [8]**
- A. Explain with example(s) the following built in functions: [Any TWO]
 - i. Round()
 - ii. COUNT(*)
 - iii. SUBSTR()
 - B. Write short note on types of SQL join.
- OR**
- Q5 Write SQL command for each of the following. [Any FOUR] [8]**
- i. Create a table EMP having fields (ENO number(3), DNO number(1), ENAME varchar2(15), BASIC_PAY (8,2)), where ENO is primary key.
 - ii. List out DNO and DNO wise total BASIC_PAY for all employees of table EMP.
 - iii. Display all the information of table EMP in descending order of BASIC_Pay.
 - iv. Delete all the records of a table EMP.
 - v. Add primary key DEPTNO to a, existing table DEPT.
- Q6 Do as directed: [8]**
- A. Define: Cursor and write the steps to manage explicit cursor.
 - B. Explain briefly the PL/SQL block structure.
- OR**
- Q6 Assume that table named IMAST has fields ITEMNO Number(5), PRICE number(7,2), NUNITS number(2) and BAMT number (11,2). [8]**

Write a PL/SQL code block to read ITEMNO from user. If ITEMNO is found in the table IMAST then calculate and update the BAMT for that ITEMNO using formula: **BAMT = NUNITS * PRICE**. If ITEMNO does not found in table IMAST then display message 'Given ITEMNO XXX does not exist in the table IMAST'.